

Geophysical Research Letters®



RESEARCH LETTER

10.1029/2026GL124747

Key Points:

- We compare daily and monthly extreme precipitation trends across five observational data sets and three Earth system model ensembles
- An unexpected number of locations have observed extreme precipitation trends that fall above or below the model distributions
- Higher model resolution does not necessarily result in better agreement with observed extreme precipitation trends

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

F. V. Davenport,
f.davenport@colostate.edu

Citation:

Davenport, F. V., & Touma, D. (2026). Discrepancies in extreme daily and monthly precipitation trends between observations and Earth system models. *Geophysical Research Letters*, 53, e2026GL124747. <https://doi.org/10.1029/2026GL124747>

Received 1 JUN 2026
Accepted 12 SEP 2026

Author Contributions:

Conceptualization: Frances V. Davenport, Danielle Touma
Data curation: Frances V. Davenport
Formal analysis: Frances V. Davenport
Methodology: Frances V. Davenport, Danielle Touma
Software: Frances V. Davenport
Visualization: Frances V. Davenport, Danielle Touma
Writing – original draft: Frances V. Davenport
Writing – review & editing: Frances V. Davenport, Danielle Touma

© 2026 The Author(s). Geophysical Research Letters published by Wiley Periodicals LLC on behalf of American Geophysical Union.
This is an open access article under the terms of the [Creative Commons Attribution License](#), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Discrepancies in Extreme Daily and Monthly Precipitation Trends Between Observations and Earth System Models

Frances V. Davenport¹  and Danielle Touma² 

¹Civil and Environmental Engineering, Colorado State University, Fort Collins, CO, USA, ²Jackson School of Geosciences, Institute for Geophysics, University of Texas, Austin, TX, USA

Abstract While extreme precipitation is expected to intensify globally in a warming climate, changes in extreme precipitation on regional scales remain uncertain. To understand how well Earth system models (ESMs) capture recent changes in regional extreme precipitation, we compare trends in daily and monthly extreme precipitation across multiple observational data sets and three ESM ensembles. Observations and ESMs show similar increases in the percent of land area with positive trends in daily precipitation extremes over time, but ESMs underestimate the magnitude of both positive and negative observed trends in daily extremes. For monthly extremes, ESMs simulate much more widespread increases compared to observations in recent decades. As a result, many recent observed decreasing trends in monthly extreme precipitation fall outside the model distribution. We find similar results when comparing observed trends to those in high-resolution simulations. These results highlight ongoing challenges in constraining the magnitude of regional extreme precipitation change.

Plain Language Summary Scientists expect heavy rainfall events to become more intense as the climate warms, but it is still difficult to predict exactly how extreme precipitation will change in specific regions. To test how well climate models capture recent changes in extreme precipitation, we compare trends from multiple historical precipitation data sets with results from three sets of climate model simulations. We find that many of the observed trends in extreme precipitation fall outside the range predicted by the models, even for high-resolution models. Our results emphasize the need for more research to understand the factors causing regional extreme precipitation change in observations and models in order to make more accurate future projections.

1. Introduction

Globally, a warmer atmosphere is expected to cause an increase in extreme precipitation, resulting from the exponentially increasing water vapor holding capacity of the atmosphere with temperature (Allen & Ingram, 2002; Trenberth, 1999). This change is supported by theory and model simulations, and numerous studies have documented observed global increases in extreme precipitation as a result of recent warming (Dong et al., 2021; Fischer & Knutti, 2016; Min et al., 2011; Papalexiou & Montanari, 2019; Zhang et al., 2013). However, the magnitude of extreme precipitation change at local-to-regional scales is much less certain (Sen-eirratne et al., 2021), which presents a challenge for attributing historical climate change impacts and informing future adaptation.

Uncertainty in regional extreme precipitation change stems from multiple sources. Extreme precipitation is highly influenced by natural variability, and detecting statistically significant changes in observations is challenging without sufficiently long historical time series (Fischer & Knutti, 2014). Large ensembles of Earth system model (ESM) simulations help overcome this challenge by providing a robust estimate of internal variability, however there is high uncertainty in simulated extreme precipitation on regional scales due to the coarse resolution of ESMs. Further, different ESMs disagree about the magnitude of regional extreme precipitation change in response to external forcing, often due to model differences in the response of the large-scale atmospheric circulation (Pfahl et al., 2017; Ritzhaupt & Maraun, 2024) or precipitation parameterizations (O’Gorman, 2012).

Recent studies have evaluated signals of climate change by comparing observed trends with those simulated by ESMs (e.g., Hausfather et al., 2020; Shaw et al., 2024; Simpson et al., 2025 and references therein). These efforts, motivated by more consistent observational coverage and a larger signal of forced change over the past few decades, have revealed both agreement and disagreement between modeled and observed changes. For example,

models generally show similar rates of global mean warming (Hausfather et al., 2020) and increases in total column water vapor (Douville et al., 2022) compared to observations. Both models and observations also show increases in precipitation variability (Zhang et al., 2024) and in the intensity of extreme precipitation (Fischer & Knutti, 2016; Zhang et al., 2013). At the same time, observed increases in daily extreme precipitation may be underestimated by models (Allan & Soden, 2008; Asadieh & Krakauer, 2015; Borodina et al., 2017; Min et al., 2011).

In this study, we compare regional extreme precipitation changes over land in observations and ESMs. To build on prior studies, we analyze both daily and longer-duration (monthly) extremes. Monthly precipitation extremes are linked to major flood impacts (e.g., Davenport et al., 2021; Hwang & Lall, 2024; Merz et al., 2021) but changes in longer-duration extremes have received less attention. We also compare standard and high-resolution ESMs to evaluate whether resolution improves trend agreement, and over multiple time periods to understand whether agreement has changed as the magnitude of external climate forcing has increased.

2. Methods

We compare linear trends in extreme precipitation across observations and ESMs for annual maximum 1-day precipitation amount (Rx1day) and the 95th percentile of cumulative monthly precipitation (monthly p95).

2.1. Historical Precipitation Data Sets

We compare four observational precipitation data sets:

- The Global Precipitation Climatology Centre (GPCC) Full Data Monthly Product version 2022 (Schneider et al., 2011). GPCC has global land coverage at $2.5^\circ \times 2.5^\circ$ resolution from 1891 to 2020, and is based on data from >52,000 in situ stations. Due to the low density of stations prior to ~1930, we only analyze the period from 1930 to 2020.
- The Global Precipitation Climatology Project (GPCP) Monthly Analysis Product (Adler et al., 2003). GPCP has global coverage at $2.5^\circ \times 2.5^\circ$ resolution from 1979–present. GPCP merges in situ, satellite and sounding observations.
- The CPC Global Unified Gauge-Based Analysis of Daily Precipitation (Chen et al., 2008). CPCP has global land coverage at $0.5^\circ \times 0.5^\circ$ resolution from 1979–present, and is based on data from >30,000 in situ stations.
- The Rainfall Estimates on a Gridded Network (REGEN) version 1–2019 Long Term stations daily precipitation product (Contractor et al., 2020). REGEN has global land coverage at $1^\circ \times 1^\circ$ resolution from 1950 to 2016. REGEN Long Term is based on stations from the Global Historical Climatology Network—Daily (GHCN-Daily) and GPCC networks that have at least 40 years of complete data.

We also analyze one precipitation reanalysis product,

- The Multi-Source Weighted-Ensemble Precipitation (MSWEP) version 2.8 (Beck et al., 2019). We analyze the monthly and daily gauge-corrected products, which have global coverage at $0.1^\circ \times 0.1^\circ$ resolution from 1979 to 2020. MSWEP combines in situ, satellite, and reanalysis data.

We collectively refer to MSWEP and the other data sets as “observations” because MSWEP incorporates observational data sets. However, MSWEPv2 likely suffers from some of the same biases as other reanalysis precipitation, such as overestimation of light precipitation and underestimation of extreme events (Beck et al., 2019; Lavers et al., 2022; Wang et al., 2026).

2.2. Earth System Model Ensembles

We analyze three ESM ensembles:

- The Coupled Model Intercomparison Project Phase 6 (CMIP6) archive (Eyring et al., 2016). We analyze one simulation each from 44 ESMs with monthly precipitation and from 37 ESMs with daily precipitation. Individual models and simulations analyzed are listed in Table S1 of Supporting Information S1. We combine historical simulations for 1930–2014 with SSP5-8.5 forcing simulations for 2015–2020.
- The 30-member GFDL SPEAR Large Ensemble (Delworth et al., 2020). The GFDL SPEAR Large ensemble simulations were run using the SPEAR_MED model with 50 km nominal resolution in the atmosphere. We combine historical simulations for the period 1930–2014 with SSP5-8.5 simulations for 2015–2020.

- A 9-member ensemble of high-resolution simulations from the MESACLIP project (Chang et al., 2020). These simulations use a high-resolution version of the Community Earth System Model (CESM) with nominal resolution of ~ 25 km in the atmosphere and ~ 4 – 10 km in the ocean. We combine historical simulations for the period 1930–2005 with RCP8.5 simulations for 2006–2020.

2.3. Extreme Precipitation Trend Calculations

We regrid all data sets to a common $5^\circ \times 5^\circ$ latitude-longitude grid using conservative regridding from the xESMF Python package (Zhuang et al., 2025). Previous work has shown that regridding precipitation to a shifted grid of similar resolution significantly alters the statistics of precipitation (Rajulapati et al., 2021). Therefore, we use $5^\circ \times 5^\circ$ resolution (which is coarser than all observations and ESMs), so that the regridding corresponds to a coarsening of all data sets rather than a shift of some data sets and not others.

For data sets with global coverage, we apply a mask to only include land grid cells before regridding. For GPCP, we use grid cells covered by the GPCC land-only data set. For most CMIP6 models, we create the land mask from model-specific land surface fraction files available through CMIP6. For CMIP6 models without land fraction files, MESACLIP and SPEAR simulations, and MSWEP, we use the 0.1° IMERG (Huffman et al., 2019) land fraction file to determine land grid cells (defined as $>5\%$ land area). We use the inverse distance weighted regridding algorithm from the UXarray Python package (UXarray Organization, 2024) to regrid the unstructured MESACLIP grid to a regular $0.25^\circ \times 0.25^\circ$ lat-lon grid before applying the land mask and regridding to the $5^\circ \times 5^\circ$ grid. In total, there are 693 $5^\circ \times 5^\circ$ land grid cells with overlapping coverage across all observational and model data sets. All percentages reported in figures and text are weighted by grid-cell area.

We mainly analyze the 42-year period from 1979 to 2020 when there is overlapping coverage across all observational data sets (except REGEN which is used for later analyses). To calculate Rx1day trends, we first calculate Rx1day in each $5^\circ \times 5^\circ$ grid cell in each year, and then calculate the linear trend in annual Rx1day over 1979–2020. We also calculate the mean Rx1day value for the 1979–2020 period, and report the trend in Rx1day as a percent change relative to this mean.

For monthly precipitation, we use quantile regression (Koenker & Bassett, 1978) to calculate the trend in the 95th percentile of monthly precipitation from 1979 to 2020 in each grid cell. Quantile regression is used to estimate the conditional quantile of a distribution, here as a function of time, and has previously been used to analyze trends in temperature or precipitation extremes (Davenport et al., 2021; McKinnon et al., 2016). We also calculate the climatological 95th percentile across all monthly values in this time period, and report the trend as a percent change relative to this value. We perform quantile regression calculations using the pyqreg Python package (Kim, 2022).

2.4. Comparing Trends in Observations vs. ESMs

We calculate the empirical quantile of each observed trend within the distribution of trends from a given model ensemble, separately for each grid cell. Observed trends that fall above (below) a given model distribution have an empirical quantile value of one (zero). Because there are many land grid cells, we expect that some observed trends lie outside of the model distribution by chance. To quantify whether the observations differ from the individual ensemble members in this respect, we add the observations as an additional ensemble member to each model distribution. We then select one ensemble member at a time and calculate the quantile of each trend from that ensemble member within the remaining distribution. We compare the total number of observed trends falling above and below the model distribution to the number of trends falling above or below the rest of the distribution for each ensemble member. For the CMIP6 and SPEAR ensembles, we also repeat these calculations but using a 9-member subset of the original ensemble each time (Figure S6 in Supporting Information S1). This allows us to compare results across models while explicitly accounting for the smaller size of the MESACLIP ensemble.

2.5. Sensitivity to Trend Time Period

We repeat the analysis above for other 42-year time periods beginning every 5 years between 1930 and 1975 using the GPCC monthly precipitation data and every 5 years between 1950 and 1975 using the REGEN daily precipitation data. We also repeat the analysis above for trends of varying length. For monthly p95 precipitation, we analyze trends starting every 5 years between 1930 and 1990 and ending in 2020, resulting in trends ranging from

31 to 91 years. We analyze Rx1day trends starting every 5 years between 1950 and 1985 and ending in 2016 (the end year for REGEN data), resulting in trends ranging from 32 to 67 years. We calculate these trends using all observational products with complete temporal coverage for a given time period. We report trend magnitudes as percent change relative to the climatological monthly p95 or Rx1day value calculated over the trend period. To minimize inconsistencies due to changes in gauge availability across time periods, we apply the quality mask associated with the REGEN LongTermStations product to subset this analysis to regions with higher-quality long-term data (Figure S1 in Supporting Information S1).

3. Results

Figures 1a and 1b show observed trends in Rx1day and monthly p95 precipitation over 1979–2020 (other data sets shown in Figure S2 of Supporting Information S1). Depending on the data set, 50%–66% of land area experienced positive Rx1day trends over this period, concentrated across Eurasia, South America, and the United States. Positive trends in monthly p95 precipitation are less widespread, with 45%–55% of area having positive trends depending on the data set. GPCC and GPCP agree on the sign of monthly p95 precipitation trends for ~72% of land area, whereas CPC and REGEN agree on the sign of Rx1day trends for ~63% of area (Figure S3 in Supporting Information S1; note that REGEN trends are based on a different time period). Agreement with MSWEP is slightly lower, with only 52% of area (monthly p95) and 40% of area (Rx1day) having consistent trends across all observational data sets.

The ensemble mean maps (Figures 1c–1h) show increases in extreme precipitation for most land regions across both metrics, although increases in Rx1day are slightly more widespread (~78–95% of area) than increases in monthly p95 precipitation (~74–84% of area). Although the forced response (i.e., ensemble-mean) indicates more widespread increases in extreme precipitation than observed, this is not an apples-to-apples comparison because the observed trends also include internal variability. The percent area with positive extreme precipitation trends in individual ensemble members ranges from ~56 to 78% for Rx1day trends (Figure 1i) and from ~51 to 78% for monthly p95 trends (Figure 1j), depending on the ESM ensemble. Overall, the percent area with positive trends is lower for MSWEP compared to other observational data sets, and slightly higher for SPEAR compared to the other ensembles. While CPC falls within the MESACLIP and CMIP6 distributions for Rx1day trends, all other observational data sets fall at the lower end or below the model distributions for both metrics.

In Figure 2, we compare the magnitude of observed trends with the CMIP6 distribution. Observed trends are split into maps for negative trends (left column) and positive trends (right column) to visualize whether trends falling above or below the model distribution are positive or negative. From these maps, it is clear that observed trends falling above the CMIP6 distribution are more positive than any individual ensemble members, and trends falling below are more negative. Depending on the observational data set, between ~6 and 19% of positive Rx1day trends fall above the entire CMIP6 distribution (Figures 2b and 2d), while ~30–40% of negative Rx1day trends fall below the CMIP6 distribution (Figures 2a and 2b). Observed Rx1day trends falling outside the model distributions occur across the globe, although there is some spatial clustering. In the CPC data, many of the Rx1day trends falling below CMIP6 occur in northern North America, Greenland, northern Europe, and central Africa. In the MSWEP data, Rx1day trends falling below CMIP6 occur mostly in central South America, central Africa, Indonesia and central Asia.

The percent area with positive monthly p95 trends falling above the CMIP6 distribution is smaller, ranging from ~8% in GPCC to ~4% in GPCP to <2% in MSWEP (Figures 2f–2h and 2j). The percent area with negative observed monthly p95 trends falling below the CMIP6 distribution ranges from 16% to 21% across the observational data sets (Figures 2e–2g and 2i). As with Rx1day trends, observed monthly p95 trends falling outside the model distribution are spread across most regions globally, with some spatial clustering depending on the observational data set and model combination. For both Rx1day and monthly p95 trends, we find similar spatial patterns but a larger number of observed trends falling outside the SPEAR and MESACLIP ensembles (Figures S4 and S5 in Supporting Information S1).

Figure 3 summarizes the percent area with observed trends falling above or below the model distribution across all observational data sets and model ensembles. In many cases, these percentages are outside the range we could expect from sampling individual ensemble members (i.e., outside the gray boxplots in Figure 3). The percentage of observed trends falling below the model distributions is greater than expected based on our resampling test for both Rx1day and monthly p95 precipitation for every observational data set and model ensemble combination, as

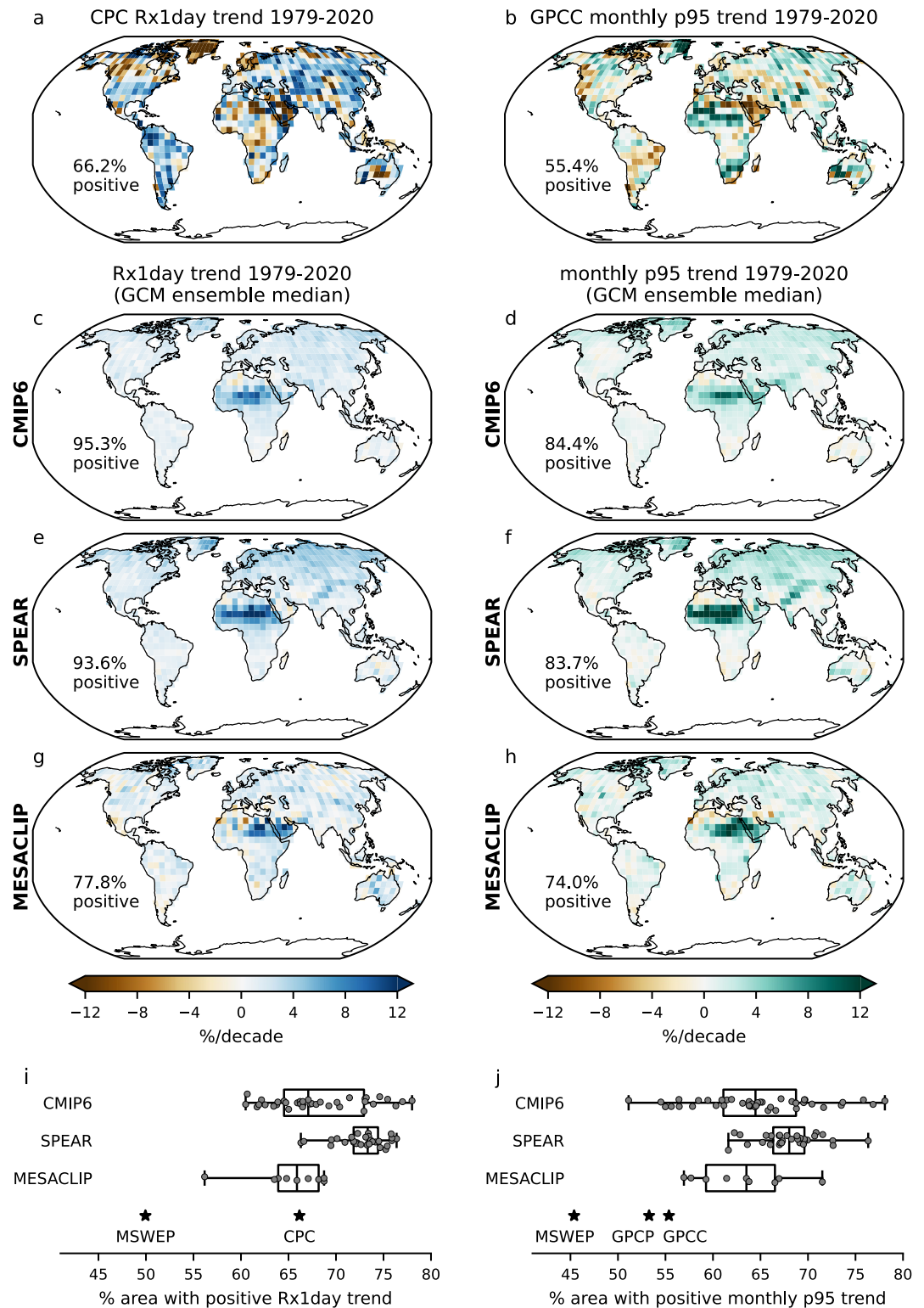
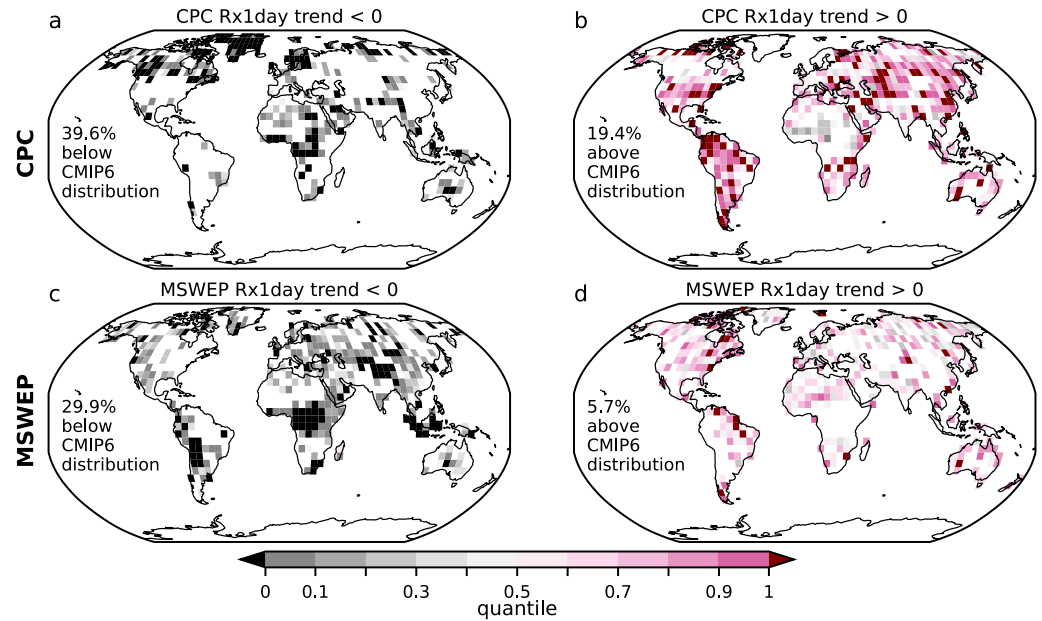


Figure 1. Trends in Rx1day (left column) and monthly p95 precipitation (right column) over 1979–2020. (a) CPC Rx1day trends. (b) GPCC monthly p95 trends. (c–h) Ensemble mean trends for CMIP6, SPEAR, and MESACLIP. (i, j) Percent of land area with positive trends over 1979–2020 in individual ensemble members and observational data sets.

Observed 1979-2020 Rx1day trends within CMIP6 trend distribution



Observed 1979-2020 monthly p95 trend within CMIP6 trend distribution

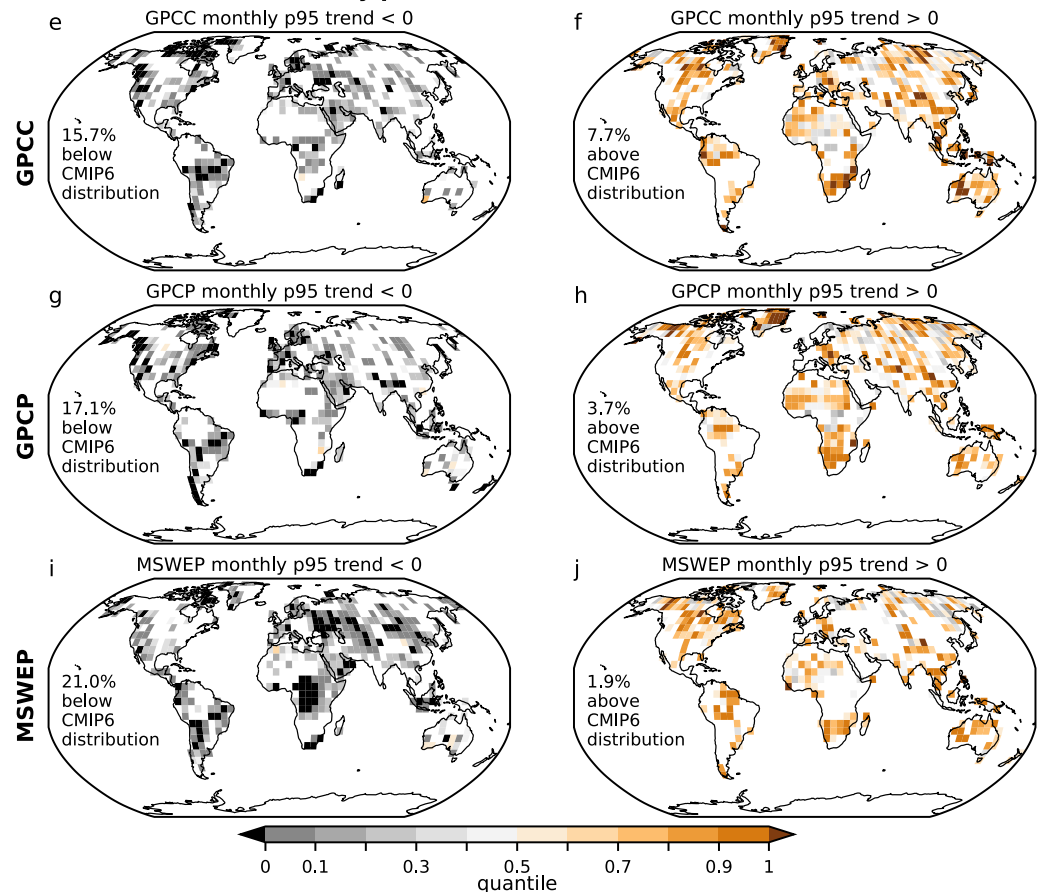


Figure 2. Quantile of observed extreme precipitation trends within the CMIP6 distribution. (a) Quantile of negative CPC Rx1day trends within CMIP6 distribution. (b) Quantile of positive CPC Rx1day trends within the CMIP6 distribution. In panels (c, d) same as (a, b) but for MSWEP Rx1day trends. In panels (e–j) same as (a, b) but for GPCC, GPCP, and MSWEP monthly p95 trends. White shading indicates omitted grid cells.

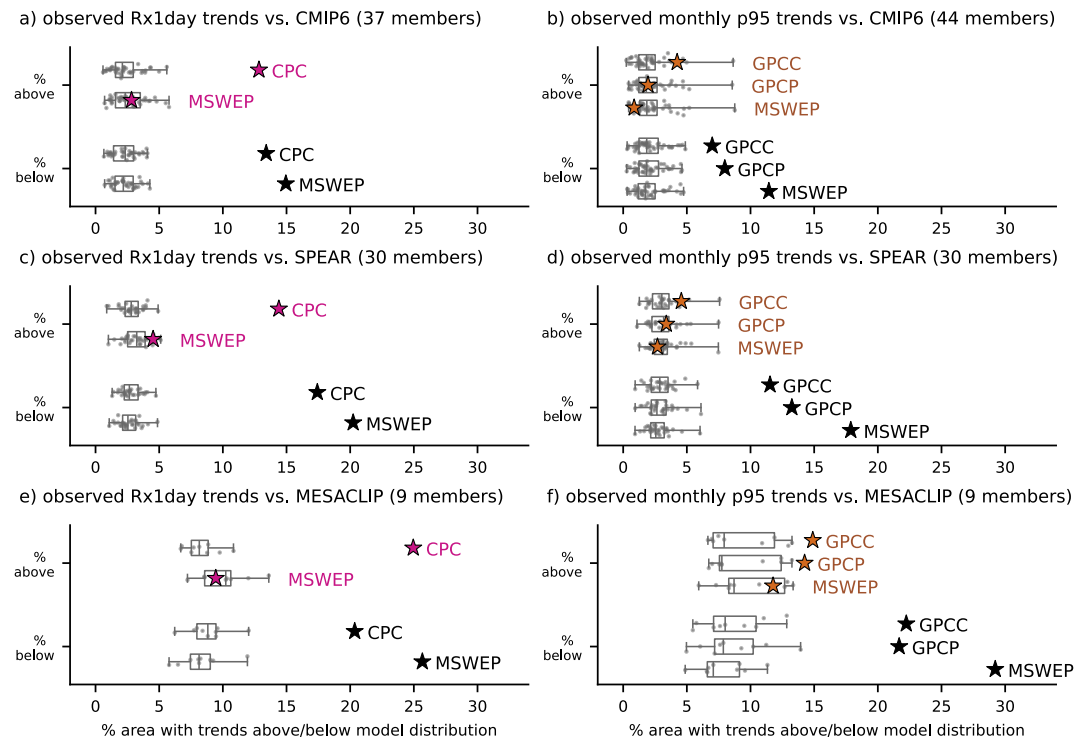


Figure 3. (a) Percent land area with observed 1979–2020 Rx1day trends falling above/below the CMIP6 distribution (stars). Gray points show results comparing individual CMIP6 ensemble members to the rest of the CMIP6 distribution (see Section 2.4). Boxplots show the min, max, 25th, 50th, and 75th percentiles across individual ensemble members. In panels (b) Same as (a) but for 1979–2020 monthly p95 trends. In panels (c–f) same as (a, b) but compared to SPEAR and MESACLIP.

is the percentage of CPC Rx1day trends falling above the model distributions. The percentage of MSWEP Rx1day trends and the percentage of monthly p95 trends that fall above the model distribution are within the expected range based on the resampling test. However, this may be for the wrong reasons, given that the overall percent area with positive trends is below or near the lower end of the model distributions for Rx1day trends in MSWEP and monthly p95 trends is all observational data sets (Figures 1i and 1j).

Figure S6 shows a version of Figure 3 where the observed trends are compared to 9-member ensembles from CMIP6 and SPEAR. These results show that differences between ensembles in Figure 3 is mostly explained by smaller ensemble sizes for SPEAR and MESACLIP, and a roughly similar number of trends fall outside all three model distributions when using a 9-member ensemble. Unlike the single-model ensembles, the multi-model CMIP6 ensemble also includes model uncertainty. We find that some CMIP6 ensemble members disagree with each other more than individual members within the SPEAR and MESACLIP ensembles (i.e., the boxplots in Figure 3; Figure S6 in Supporting Information S1 have a wider spread for CMIP6 than the other ensembles), likely due to structural differences across CMIP6 models.

To explore spatial patterns in observation-model agreement, we calculate climatological aridity for each grid cell using the TerraClimate data set (Abatzoglou et al., 2018) and analyze the percent area with observed trends falling above or below the model distributions in aridity zones and by 30-degree latitude bands. We find that observed trends fall outside the model distributions at mostly similar rates across all latitude bands and aridity zones (Figures S7 and S8 in Supporting Information S1).

Figure 4 compares agreement between observed and modeled trends over time for data sets with longer temporal coverage (REGEN and GPCC) and locations with higher quality observational coverage (see Section 2.5 for details). We find that the percent area with positive Rx1day trends has risen over time, and is similar between the observations and CMIP6 for all time periods (Figures 4a and 4e).

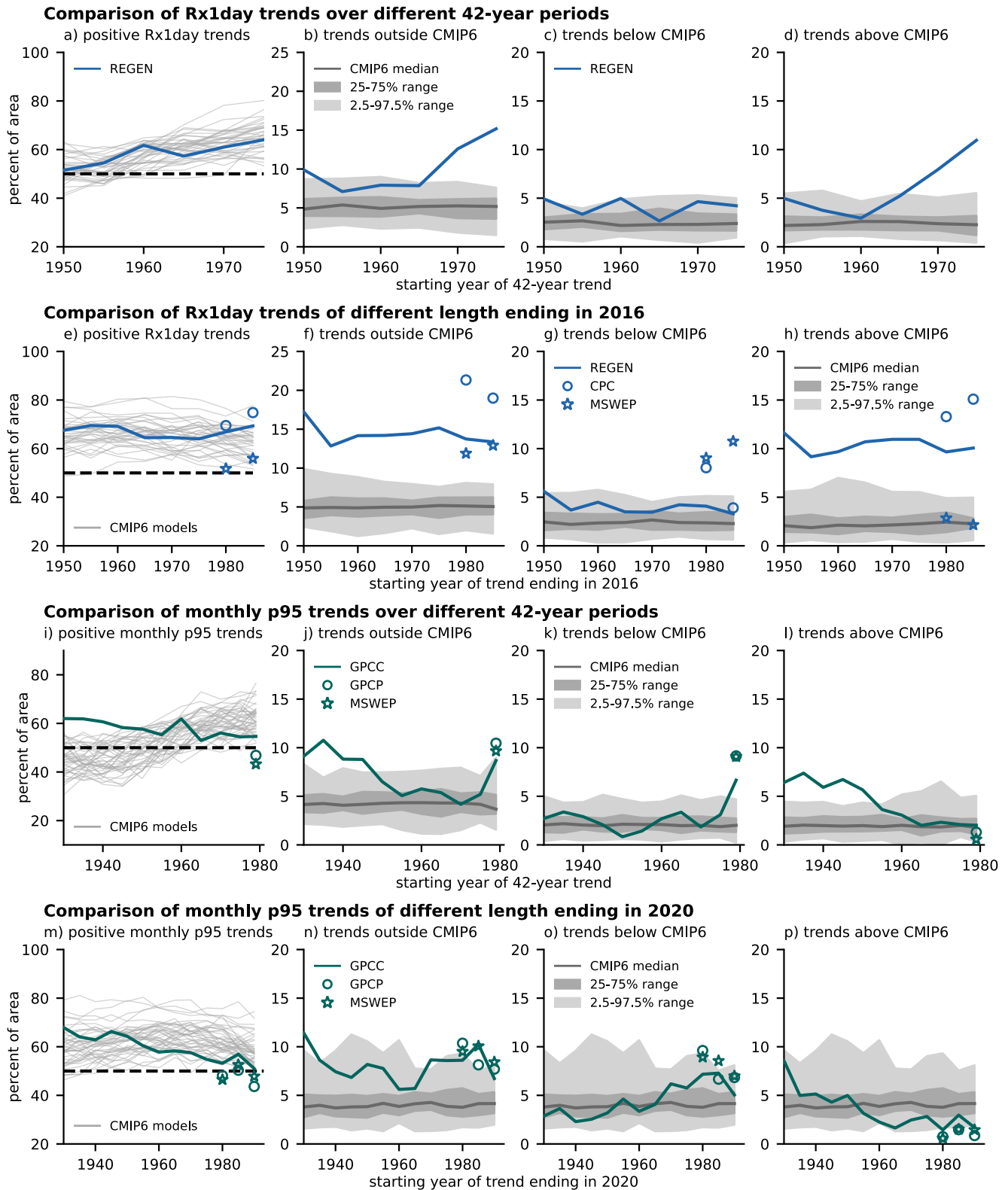


Figure 4.

When comparing the magnitude of 42-year Rx1day trends between REGEN and CMIP6, we find that percent of observed Rx1day trends falling above the CMIP6 distribution has risen in recent decades (Figure 4d). The percentage of Rx1day trends falling below the CMIP6 distribution is consistent over time (Figure 4c), although that this percentage is lower for REGEN than for CPC. We find that longer (>70-year) trends ending in 2016 show similar agreement or disagreement with models compared to shorter (<40-year) trends (Figures 4e–4h). Comparing observed REGEN Rx1day trends to SPEAR and MESACLIP over time gives similar conclusions (Figures S9 and S10 in Supporting Information S1).

The percent area with positive 42-year monthly p95 trends increases over time in all models, suggesting a strong simulated response of extreme monthly precipitation to external forcing (Figure 4i; Figures S9 and S10 in Supporting Information S1). The observations do not show this pattern; the percent area with positive 42-year monthly p95 trends in GPCC is roughly constant over time and is at the low end or below the model spread for the most recent periods (Figure 4i). Maps of 42-year GPCC monthly p95 trends for different time periods show that the magnitude of trends is consistent over time (Figure S11 in Supporting Information S1), while the number of GPCC monthly p95 trends falling below the model distribution increases over time (Figures 4k and 4o; Figures S9 and S10 in Supporting Information S1).

Results for longer (>70-year) observed monthly p95 trends are somewhat different; 60%–70% of these longer monthly p95 trends are positive (Figure 4m) and there is a higher percentage of longer GPCC monthly p95 trends falling above the model distributions compared to more recent trends (Figure 4p; Figures S9 and S10 in Supporting Information S1). Overall, increases in extreme monthly precipitation are more apparent in observations on longer timescales compared to shorter timescales. Maps of longer GPCC monthly p95 trends also show more similarities to the ensemble mean forced response compared to shorter trends (Figure S12 in Supporting Information S1).

4. Discussion and Conclusions

In this study, we compare extreme daily and monthly precipitation trends across multiple observational data sets and three ESM ensembles. While we find some areas of agreement between observed and modeled extreme precipitation trends, there are also notable discrepancies. We find that observations and models show similar increases in the percent area with positive Rx1day trends over time. However, models underestimate the magnitude of both positive and negative observed Rx1day trends. ESMs simulate much more widespread increases in monthly p95 precipitation compared to observations, at least in recent decades. As a result, many recent observed trends in monthly p95 precipitation fall below the model distributions.

These results are consistent with previous analyses comparing extreme precipitation trends between observations and models. Donat et al. (2023) compared trends in annual maximum 5-day precipitation (Rx5day) in REGEN and CMIP6 over 1950–2016, and found that CMIP6 did not capture observed regional decreases in Rx5day. Earlier studies also indicate that observed increases in extreme daily precipitation are occurring faster than models suggest (e.g., Asadieh & Krakauer, 2015; Borodina et al., 2017; Min et al., 2011). Our work indicates that these same issues may be present in high-resolution simulations, even though these simulations better capture the spatial distribution and intensity of daily extreme precipitation and project a larger increase in future extreme precipitation (Chang et al., 2026). However, our analysis is performed on a relatively coarse $5^\circ \times 5^\circ$ grid. In the future, comparing high-resolution model simulations with high-resolution observations at finer spatial resolution may yield different results.

As outlined in Simpson et al., 2025, disagreement between observed and modeled trends could have multiple sources, including an erroneous simulated forced response in models, misrepresentation of internal variability in models, biased external forcings prescribed in models, observational uncertainty or biases, or issues with the

Figure 4. (a) Percent land area with positive 42-year Rx1day trends over time in REGEN (blue) and CMIP6 ensemble members (gray lines). (b) Percent area with REGEN Rx1day trends falling outside CMIP6 distribution over time (blue line). Gray shading shows results comparing CMIP6 ensemble members with the rest of the distribution. Panels (c, d) same as (b) for REGEN Rx1day trends falling below and above the CMIP6 distribution, respectively. In panels (e–h) same as (a–d) but for trends of varying length ending in 2016. CPC and MSWEP are shown when temporal coverage overlaps. In panels (i–l) same as (a–d) but for 42-year monthly p95 trends in GPCC (green) and CMIP6 (gray), with GPCP and MSWEP shown for overlapping periods. In panels (m–p) same as (i–l) but for trends of varying length ending in 2020.

comparison method itself. Errors in aerosol emissions in models, for example, have previously been found to affect regional extreme precipitation trends (Dong et al., 2022; Lin et al., 2018; Wu et al., 2013). However, we find that discrepancies occur similarly across regions, suggesting that aerosol emission biases are likely not a primary source of disagreement here.

The fact that discrepancies in extreme precipitation trends have emerged or become worse over time (Figure 4; Figures S9 and S10 in Supporting Information S1) suggests that an erroneous forced response or in the magnitude of the forced response relative to simulated internal variability may be partly to blame. Globally, extreme precipitation in models has been shown to increase close to the 7% per °C rate expected from thermodynamics (Douvillé et al., 2021; Zhang et al., 2013). However, regional changes above or below this rate can occur due to the additional influence of large-scale circulation dynamics (Pfahl et al., 2017). Storm-scale dynamics, which are not well-represented in ESMs, could also lead to larger increases in extreme precipitation in observations compared to models, especially for shorter duration (e.g., sub-daily) extremes associated with mesoscale convective systems (Da Silva & Haerter, 2025; Gu et al., 2023; Núñez Ocasio et al., 2026). Therefore, errors in the simulated magnitude of regional extreme precipitation trends could reflect model biases in the response of both large-scale or storm-scale dynamics under warming.

Compared to daily extremes, monthly precipitation extremes might be more influenced by sea surface temperature (SST) patterns or other large-scale modes of climate variability. Prior studies have shown that ESMs likely underestimate regional climate variability, especially on decadal timescales or longer (Laepplé et al., 2023; Laepplé & Huybers, 2014). Other studies have shown that ESMs fail to simulate recent observed SST patterns, especially in the tropical Pacific (Seager et al., 2022; Wills et al., 2022). Underestimation of large-scale internal variability in models, or model errors simulating recent SST patterns and their teleconnections to land-falling precipitation could thus contribute to more disagreement for monthly p95 trends compared to Rx1day trends.

Uncertainty or biases in precipitation data sets could also be a source of disagreement with models. Observational data sets typically agree on the sign of historical trends, especially monthly p95 trends, although this is not true everywhere (Figure S3 in Supporting Information S1). Compared to other observational data sets, MSWEP has fewer increasing trends in extreme precipitation in recent decades (Figures 1i and 1j). The earliest precipitation observations are likely associated with higher uncertainty, but this is difficult to quantify given the scarcity of data sets with coverage before 1980. Future efforts to diagnose sources of discrepancies in extreme precipitation trends could benefit from including even more observational data sets, including higher-resolution data sets such as CHIRPS (Funk et al., 2026) or high-quality regional precipitation data sets.

This study adds to a growing body of literature highlighting successes and challenges in the ability of ESMs to capture recent observed Earth system changes. Constraining the magnitude of regional-to-local extreme precipitation change remains a key step for accurate climate change attribution and adaptation, and our results emphasize the need for additional research to disentangle the factors that cause extreme precipitation change in both observations and ESMs.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All data used in the analysis is publicly available. The GPCC v2022 monthly precipitation data set is available from the German Meteorological Service, Deutscher Wetterdienst (https://opendata.dwd.de/climate_environment/GPCC/html/fulldata-monthly_v2022_doi_download.html). The GPCP monthly precipitation data set is available from NOAA PSL (<https://psl.noaa.gov/data/gridded/data.gpcp.html>). The MSWEP v2.8 monthly and daily precipitation products are available from GloH2O (www.gloh2o.org) under a CC BY-NC 4.0 license. The CPC Global Unified Gauge-Based Analysis of Daily Precipitation product is provided by NOAA PSL, Boulder, CO, USA from their website (<https://psl.noaa.gov/data/gridded/data.cpc.globalprecip.html>). The REGEN Long Term daily precipitation product is available at <https://doi.org/10.25914/5ca4c2c6527d2> (Contractor et al., 2019). CMIP6 simulations are available through the Earth System Grid Federation (ESGF) data portal. Information about accessing ESGF nodes can be found at <https://wcrp-cmip.org/cmip-data-access/>. The SPEAR Large

Ensemble is available from GFDL (<https://www.gfdl.noaa.gov/spear/>) and was downloaded using nomads.gfdl.noaa.gov. The MESACLIP simulations are available through the NSF NCAR Research Data Archive: <https://project.cgd.ucar.edu/projects/MESACLIP/> (Castruccio et al., 2024a, 2024b). Research code supporting the results is available at <https://doi.org/10.5281/zenodo.22908325> (Davenport, 2026).

Acknowledgments

We thank the Earth system modeling groups and funding agencies contributing to the CMIP6, GFDL SPEAR, and MESACLIP efforts for producing and making available model simulations used in this analysis. The analysis was performed on the Colorado State University Walter Scott Jr. College of Engineering high-performance computer, Cashew.

References

- Abatzoglou, J. T., Dobrowski, S. Z., Parks, S. A., & Hegewisch, K. C. (2018). TerraClimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958–2015. *Scientific Data*, 5(1), 170191. <https://doi.org/10.1038/sdata.2017.191>
- Adler, R. F., Huffman, G. J., Chang, A., Ferraro, R., Xie, P.-P., Janowiak, J., et al. (2003). The version 2 global precipitation climatology project (GPCP) monthly precipitation analysis (1979–Present) [Dataset]. *Journal of Hydrometeorology*. [https://doi.org/10.1175/1525-7541\(2003\)004%3C1147:TVGPCP%3E2.0.CO;2](https://doi.org/10.1175/1525-7541(2003)004%3C1147:TVGPCP%3E2.0.CO;2)
- Allan, R. P., & Soden, B. J. (2008). Atmospheric warming and the amplification of precipitation extremes. *Science*, 321(5895), 1481–1484. <https://doi.org/10.1126/science.1160787>
- Allen, M. R., & Ingram, W. J. (2002). Constraints on future changes in climate and the hydrologic cycle. *Nature*, 419(6903), 224–232. <https://doi.org/10.1038/nature01092>
- Asadieh, B., & Krakauer, N. Y. (2015). Global trends in extreme precipitation: Climate models versus observations. *Hydrology and Earth System Sciences*, 19(2), 877–891. <https://doi.org/10.5194/hess-19-877-2015>
- Beck, H. E., Wood, E. F., Pan, M., Fisher, C. K., Miralles, D. G., van Dijk, A. I. J. M., et al. (2019). MSWEP V2 global 3-Hourly 0.1° precipitation: Methodology and quantitative assessment. *Bulletin of the American Meteorological Society*, 100(3), 473–500. <https://doi.org/10.1175/BAMS-D-17-0138.1>
- Borodina, A., Fischer, E. M., & Knutti, R. (2017). Models are likely to underestimate increase in heavy rainfall in the extratropical regions with high rainfall intensity. *Geophysical Research Letters*, 44(14), 7401–7409. <https://doi.org/10.1002/2017GL074530>
- Castruccio, F., Chang, P., Danabasoglu, G., Fu, D., Rosenbloom, N., Zhang, Q., et al. (2024a). MESACLIP: A 10-member ensemble of CESM HR historical (1920–2005) simulations [Dataset]. *NSF National Center for Atmospheric Research*. <https://doi.org/10.5065/7N1X-K278>
- Castruccio, F., Chang, P., Danabasoglu, G., Fu, D., Rosenbloom, N., Zhang, Q., et al. (2024b). MESACLIP: A 10-member ensemble of CESM HR RCP 8.5 (2006–2100) simulations [Dataset]. *NSF National Center for Atmospheric Research*. <https://doi.org/10.5065/PNCR-5S34>
- Chang, P., Fu, D., Liu, X., Castruccio, F. S., Prein, A. F., Danabasoglu, G., et al. (2026). Future extreme precipitation amplified by intensified mesoscale moisture convergence. *Nature Geoscience*, 19(1), 33–41. <https://doi.org/10.1038/s41561-025-01859-1>
- Chang, P., Zhang, S., Danabasoglu, G., Yeager, S. G., Fu, H., Wang, H., et al. (2020). An unprecedented set of high-resolution Earth system simulations for understanding multiscale interactions in climate variability and change. *Journal of Advances in Modeling Earth Systems*, 12(12), e2020MS002298. <https://doi.org/10.1029/2020ms002298>
- Chen, M., Shi, W., Xie, P., Silva, V. B. S., Kousky, V. E., Wayne Higgins, R., & Janowiak, J. E. (2008). Assessing objective techniques for gauge-based analyses of global daily precipitation. *Journal of Geophysical Research*, 113(D4), 2007JD009132. <https://doi.org/10.1029/2007JD009132>
- Contractor, S., Donat, M. G., Alexander, L. V., Ziese, M., Meyer-Christoffer, A., Schneider, U., et al. (2019). Rainfall estimates on a gridded network based on long-term station data v1-2019 [Dataset]. *NCI National Research Data Collection*. <https://doi.org/10.25914/5ca4c2c6527d2>
- Contractor, S., Donat, M. G., Alexander, L. V., Ziese, M., Meyer-Christoffer, A., Schneider, U., et al. (2020). Rainfall estimates on a gridded network (REGEN) – a global land-based gridded dataset of daily precipitation from 1950 to 2016. *Hydrology and Earth System Sciences*, 24(2), 919–943. <https://doi.org/10.5194/hess-24-919-2020>
- Da Silva, N. A., & Haerter, J. O. (2025). Super-Clausius–Clapeyron scaling of extreme precipitation explained by shift from stratiform to convective rain type. *Nature Geoscience*, 18(5), 382–388. <https://doi.org/10.1038/s41561-025-01686-4>
- Davenport, F. V. (2026). fdavenport/extreme-precip-trends: (version 1.1.0) [Software]. *Zenodo*. <https://doi.org/10.5281/zenodo.22908326>
- Davenport, F. V., Burke, M., & Diffenbaugh, N. S. (2021). Contribution of historical precipitation change to US flood damages. *Proceedings of the National Academy of Sciences*, 118(4), e2017524118. <https://doi.org/10.1073/pnas.2017524118>
- Delworth, T. L., Cooke, W. F., Adcroft, A., Bushuk, M., Chen, J., Dunne, K. A., et al. (2020). SPEAR: The next generation GFDL modeling system for seasonal to multidecadal prediction and projection. *Journal of Advances in Modeling Earth Systems*, 12(3), e2019MS001895. <https://doi.org/10.1029/2019MS001895>
- Donat, M. G., Delgado-Torres, C., De Luca, P., Mahmood, R., Ortega, P., & Doblas-Reyes, F. J. (2023). How credibly do CMIP6 simulations capture historical mean and extreme precipitation changes? *Geophysical Research Letters*, 50(14), e2022GL102466. <https://doi.org/10.1029/2022GL102466>
- Dong, B., Sutton, R. T., Shaffrey, L., & Harvey, B. (2022). Recent decadal weakening of the summer Eurasian westerly jet attributable to anthropogenic aerosol emissions. *Nature Communications*, 13(1), 1148. <https://doi.org/10.1038/s41467-022-28816-5>
- Dong, S., Sun, Y., Li, C., Zhang, X., Min, S.-K., & Kim, Y.-H. (2021). Attribution of extreme precipitation with updated observations and CMIP6 simulations. *Journal of Climate*, 34(3), 871–881. <https://doi.org/10.1175/JCLI-D-19-1017.1>
- Douville, H., Qasbi, S., Ribes, A., & Bock, O. (2022). Global warming at near-constant tropospheric relative humidity is supported by observations. *Communications Earth & Environment*, 3(1), 237. <https://doi.org/10.1038/s43247-022-00561-z>
- Douville, H., Raghavan, K., Renwick, J., Allan, R. P., Arias, P. A., Barlow, M., et al. (2021). Water cycle changes. In *Climate change 2021 – The physical science basis: Working group I contribution to the sixth assessment report of the intergovernmental panel on climate change* (pp. 1055–1210). Cambridge University Press. <https://doi.org/10.1017/9781009157896>
- Eyring, V., Bony, S., Meehl, G. A., Senior, C. A., Stevens, B., Stouffer, R. J., & Taylor, K. E. (2016). Overview of the coupled model inter-comparison project phase 6 (CMIP6) experimental design and organization. *Geoscientific Model Development*, 9(5), 1937–1958. <https://doi.org/10.5194/gmd-9-1937-2016>
- Fischer, E. M., & Knutti, R. (2014). Detection of spatially aggregated changes in temperature and precipitation extremes. *Geophysical Research Letters*, 41(2), 547–554. <https://doi.org/10.1002/2013GL058499>
- Fischer, E. M., & Knutti, R. (2016). Observed heavy precipitation increase confirms theory and early models. *Nature Climate Change*, 6(11), 986–991. <https://doi.org/10.1038/nclimate3110>
- Funk, C., Peterson, P., Harrison, L., Saldivar, R., Landsfeld, M., Pedreros, D., et al. (2026). The climate hazards center infrared precipitation with stations, version 3. *Scientific Data*, 13(1), 718. <https://doi.org/10.1038/s41597-026-07096-4>

- Gu, L., Yin, J., Gentile, P., Wang, H.-M., Slater, L. J., Sullivan, S. C., et al. (2023). Large anomalies in future extreme precipitation sensitivity driven by atmospheric dynamics. *Nature Communications*, 14(1), 3197. <https://doi.org/10.1038/s41467-023-39039-7>
- Hausfather, Z., Drake, H. F., Abbott, T., & Schmidt, G. A. (2020). Evaluating the performance of past climate model projections. *Geophysical Research Letters*, 47(1), e2019GL085378. <https://doi.org/10.1029/2019GL085378>
- Huffman, G. J., Bolvin, D. T., Braithwaite, D., Hsu, K., Joyce, R., Kidd, C., et al. (2019). NASA global precipitation measurement (GPM) integrated Multi-satellitE retrievals for GPM (IMERG) (NASA algorithm theoretical basis doc.). Retrieved from https://gpm.nasa.gov/sites/default/files/document_files/IMERG_ATBD_V06.pdf
- Hwang, J., & Lall, U. (2024). Increasing dam failure risk in the USA due to compound rainfall clusters as climate changes. *Npj Natural Hazards*, 1(1), 27. <https://doi.org/10.1038/s44304-024-00027-6>
- Kim, J. (2022). Pyqreg (version 0.4.2) [Software].
- Koenker, R., & Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33. <https://doi.org/10.2307/1913643>
- Laepfle, T., & Huybers, P. (2014). Ocean surface temperature variability: Large model–data differences at decadal and longer periods. *Proceedings of the National Academy of Sciences*, 111(47), 16682–16687. <https://doi.org/10.1073/pnas.1412077111>
- Laepfle, T., Ziegler, E., Weitzel, N., Hébert, R., Ellerhoff, B., Schoch, P., et al. (2023). Regional but not global temperature variability underestimated by climate models at supradecadal timescales. *Nature Geoscience*, 16(11), 958–966. <https://doi.org/10.1038/s41561-023-01299-9>
- Lavers, D. A., Simmons, A., Vamborg, F., & Rodwell, M. J. (2022). An evaluation of ERA5 precipitation for climate monitoring. *Quarterly Journal of the Royal Meteorological Society*, 148(748), 3152–3165. <https://doi.org/10.1002/qj.4351>
- Lin, L., Wang, Z., Xu, Y., Fu, Q., & Dong, W. (2018). Larger sensitivity of precipitation extremes to aerosol than greenhouse gas forcing in CMIP5 models. *Journal of Geophysical Research: Atmospheres*, 123(15), 8062–8073. <https://doi.org/10.1029/2018JD028821>
- McKinnon, K. A., Rhines, A., Tingley, M. P., & Huybers, P. (2016). The changing shape of Northern Hemisphere summer temperature distributions. *Journal of Geophysical Research: Atmospheres*, 121(15), 8849–8868. <https://doi.org/10.1002/2016JD025292>
- Merz, B., Blöschl, G., Vorogushyn, S., Dottori, F., Aerts, J. C. J. H., Bates, P., et al. (2021). Causes, impacts and patterns of disastrous river floods. *Nature Reviews Earth & Environment*, 2(9), 592–609. <https://doi.org/10.1038/s43017-021-00195-3>
- Min, S.-K., Zhang, X., Zwiers, F. W., & Hegerl, G. C. (2011). Human contribution to more-intense precipitation extremes. *Nature*, 470(7334), 378–381. <https://doi.org/10.1038/nature09763>
- Núñez Ocasio, K. M., Dougherty, E. M., & Moon, Z. L. (2026). Mesoscale convective systems in Africa and the Atlantic Ocean under a kilometer-scale future climate scenario. *Monthly Weather Review*, 154(8), 1573–1589. <https://doi.org/10.1175/MWR-D-25-0207.1>
- O’Gorman, P. A. (2012). Sensitivity of tropical precipitation extremes to climate change. *Nature Geoscience*, 5(10), 697–700. <https://doi.org/10.1038/ngeo1568>
- Papalexio, S. M., & Montanari, A. (2019). Global and regional increase of precipitation extremes under global warming. *Water Resources Research*, 55(6), 4901–4914. <https://doi.org/10.1029/2018wr024067>
- Pfahl, S., O’Gorman, P. A., & Fischer, E. M. (2017). Understanding the regional pattern of projected future changes in extreme precipitation. *Nature Climate Change*, 7(6), 423–427. <https://doi.org/10.1038/nclimate3287>
- Rajulapati, C. R., Papalexio, S. M., Clark, M. P., & Pomeroy, J. W. (2021). The perils of regridding: Examples using a global precipitation dataset. *Journal of Applied Meteorology and Climatology*. <https://doi.org/10.1175/JAMC-D-20-0259.1>
- Ritzhaupt, N., & Maraun, D. (2024). State-of-the-art climate models reduce dominant dynamical uncertainty in projections of extreme precipitation. *Environmental Research: Climate*, 3(2), 021001. <https://doi.org/10.1088/2752-5295/ad2eb2>
- Schneider, U., Becker, A., Finger, P., Meyer-Christoffer, A., Rudolf, B., & Ziese, M. (2011). GPCC full data reanalysis version 6.0 at 2.5°: Monthly land-surface precipitation from rain-gauges built on GTS-Based and Historic data [Dataset]. https://doi.org/10.5676/DWD_GPCC/FD_M_V7_250
- Seager, R., Henderson, N., & Cane, M. (2022). Persistent discrepancies between observed and modeled trends in the tropical Pacific Ocean. *Journal of Climate*, 35(14), 4571–4584. <https://doi.org/10.1175/JCLI-D-21-0648.1>
- Seneviratne, S. I., Zhang, X., Adnan, M., Badi, W., Di Luca, A., Ghosh, S., et al. (2021). Weather and climate extreme events in a changing climate. In *Climate change 2021: The physical science basis. Contribution of working group I to the sixth assessment report of the intergovernmental panel on climate change* (pp. 1513–1766). Cambridge University Press. <https://doi.org/10.1017/9781009157896.013>
- Shaw, T. A., Arias, P. A., Collins, M., Coumou, D., Diedhiou, A., Garfinkel, C. I., et al. (2024). Regional climate change: Consensus, discrepancies, and ways forward. *Frontiers in Climate*, 6, 1391634. <https://doi.org/10.3389/fclim.2024.1391634>
- Simpson, I. R., Shaw, T. A., Ceppi, P., Clement, A. C., Fischer, E., Grise, K. M., et al. (2025). Confronting Earth system model trends with observations. *Science Advances*, 11(11), eadt8035. <https://doi.org/10.1126/sciadv.adt8035>
- Trenberth, K. E. (1999). Conceptual framework for changes of extremes of the hydrological cycle with climate change. *Climatic Change*, 42(1), 327–339. <https://doi.org/10.1023/A:1005488920935>
- UXarray Organization. (2024). UXarray (version 2024.11.1) [Software]. <https://doi.org/10.5281/zenodo.5655065>
- Wang, X., Alharbi, R. S., Baez-Villanueva, O. M., Miralles, D. G., Ma, J., Xu, S., et al. (2026). Mswep V3: Machine learning-powered global precipitation estimates at 0.1° hourly resolution (1979–Present). *arXiv*. <https://doi.org/10.48550/arXiv.2602.01436>
- Wills, R. C. J., Dong, Y., Proistosescu, C., Armour, K. C., & Battisti, D. S. (2022). Systematic climate model biases in the large-scale patterns of recent sea-surface temperature and sea-level pressure change. *Geophysical Research Letters*, 49(17), e2022GL100011. <https://doi.org/10.1029/2022GL100011>
- Wu, P., Christidis, N., & Stott, P. (2013). Anthropogenic impact on Earth’s hydrological cycle. *Nature Climate Change*, 3(9), 807–810. <https://doi.org/10.1038/nclimate1932>
- Zhang, W., Zhou, T., & Wu, P. (2024). Anthropogenic amplification of precipitation variability over the past century. *Science*, 385(6707), 427–432. <https://doi.org/10.1126/science.adp0212>
- Zhang, X., Wan, H., Zwiers, F. W., Hegerl, G. C., & Min, S. (2013). Attributing intensification of precipitation extremes to human influence. *Geophysical Research Letters*, 40(19), 5252–5257. <https://doi.org/10.1002/grl.51010>
- Zhuang, J., Dussin, R., Bourgault, P., Huard, D., Banihirwe, A., Raynaud, S., et al. (2025). xESMF (version 0.8.10) [Software]. <https://doi.org/10.5281/zenodo.4294774>